2D Hector SLAM of Indoor Mobile Robot using 2D Lidar

Shubham Nagla

Electronics and Communication Engineering

Guru Nanak Dev Engineering College

Ludhiana, Punjab, India

Shubhamnagla6280@gmail.com

Abstract*—Laser Range Finders are being widely used in SLAM research. In this paper, a 2D-SLAM algorithm based on LiDAR in the robot operating system (ROS) is evaluated, the name for the same is Hector-SLAM. In order to reflect the ability of building maps by using Hector-SLAM algorithm, experiments were carried out in a custom build L shaped environment. The map was created in real time and we can also observe the path that was followed by the mobile robot during the whole process. Finally, a 2D map of the L shaped environment was further saved in the system which can be further used for various applications.

Keywords*—Robot Operating System (ROS), Simultaneous Localization and Mapping (SLAM), Hector-SLAM, Light Detection and Ranging (LiDAR).

I. INTRODUCTION

Simultaneous Localization and Mapping (SLAM) of mobile robots in unknown environment is prevailing in robotics research. SLAM is basically mapping an unknown environment and while performing that we also track the location of the object moving in the certain environments. We can perform SLAM on the basis of the sensor being used such as by laser sensor using LiDAR or by using visual sensors such as stereo cameras. The map of environment is required for a mobile robot to manoeuvre from room to room, picking up objects from one place and placing it to a different one. To perform such actions, the robot must not only be aware about the environment but also of its own location in that environment [1].

The robot used is a simple wired mobile robot being controlled by the user to move inside the environment. The robot is installed with a Hokuyo LiDAR on the front of it. Which is being used to perform the Hector-SLAM.

Robot Operating System (ROS) is being used for running the algorithms to achieve SLAM. At the end we are able to visualize the whole process in the software called RViz. The map is further saved in the system.

Figure1: Block Diagram of the System.

A. Robot Operating System (ROS):

ROS (Robot Operating System) provides libraries and tools to create robot applications. It provides hardware abstraction, device drivers, libraries, visualizers, message-passing, package management, and more. It is also used as an operating system which is used to perform Mapping, localization, and autonomous navigation in any unknown environment are popular problems in the field of autonomous mobile robots [2]. In the proposed project we are using Melodic distribution of ROS on a Linux machine which is connected to the LiDAR and further the data that is being received from LiDAR is used to make a 2D Map of the environment.

Advantages of ROS:

- Makes it easy to get the innovations from the university labs out into the market
- User testing will find and patch bugs quickly
- Users can always add or change the source code to suit their own purposes
- Open-source, BSD licensed code makes it easy for users to incorporate ROS functionality into their products

B. Light Detection and Ranging (LiDAR)

It is a method for estimating the distances (ranging) by hitting the target with a beam of laser and measuring the reflection with a sensor. Further that data is used to perform various tasks like 2D mapping, 3D mapping, 3D modelling and etc.

LiDAR's high resolution and accuracy has enhanced it to be used in the creation of maps. Is also used

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along with aerial photography, LiDAR can also be uses in road, building and vegetation mapping.

The 3D and 2D aspects of LiDAR makes it perfect for 2D and 3D mapping, including complex mountain topography.

We are using the Hokuyo LiDAR for performing the 2D mapping in the unknown L shaped environment. By mounting it on to a mobile robot which is made to move inside an unknown environment.

We can further visualize the same on RViz along with the path highlighted with green coloured line.

Figure2: Hokuyo Rapid LiDAR

C. Hector-SLAM:

Hector-SLAM algorithm uses high resolution and high frequency laser data and ignores the odometer information. The method used for performing mapping for an unknown environment in this algorithm is grid mapping and is located by scanning matching. At the home position, the first frame data being provided by LiDAR is directly utilized in building a graph, then the sensor data is compared with the map and an optimal pose of the LiDAR unit is derived [3].

The optimal pose of the LiDAR unit is then given to the hector mapping node running on ROS. Which registers the optimal position of the LiDAR unit as the base link and odometry frame for the algorithm. So, that we don't have to provide the odometry information.

The optimal solution is searched and conducted by using the Gauss-Newton gradient method, as shown below:

Let x_t be the optimal pose of the lidar.

$$x_t = \arg min \sum_{i=1}^{n} [1 - M(S_i(x_t))]^2 \quad (1)$$

Where $S_i(x_t)$ represent the coordinates $S_i = (S_{ix}, S_{iy})$ of the endpoint coordinate of the lidar in the global coordinate system, as shown in equation (2)

$$S_i(x_t) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} S_{ix} \\ S_{iy} \end{bmatrix} + \begin{bmatrix} x \\ y \end{bmatrix} \quad (2)$$

$M(S_i(x_t))$ represents grid map value when the coordinate is $S_i(x_t)$. Following the idea of the Gauss-Newton gradient method [4], an initial estimated value $\overline{x}_t$ is given according to the equation (3) to estimate Δx_t:

$$\sum_{i=1}^{n} [1 - M(S_i(\overline{x_t} + \Delta x_t))]^2 \qquad (3)$$

Expanding $1 - M(S_i(\overline{x_t} + \Delta x_t))$ with a first-order Taylor expansion, and obtaining a value of 0 for a partial derivative of Δx_t, we can get:

$$\Delta x_t = H^{-1} \sum_{i=1}^{n} \left[\nabla M(S_i(\overline{x_t})) \frac{\partial S_i(\overline{x_t})}{\partial \overline{x}_t} \right] [1 - M(S_i(\overline{x_t}))]$$

And

$$H = \left[\nabla M(S_i(\overline{x_t})) \frac{\partial S_i(\overline{x_t})}{\partial \overline{x}_t} \right]^T \left[\nabla M(S_i(\overline{x_t})) \frac{\partial S_i(\overline{x_t})}{\partial \overline{x}_t} \right]$$

To generate better map, we need to use the method of loop closure detection but in Hector SLAM we particularly don't need that and still it can accurately complete the loop in many real scenes.

II. Hardware System and Architecture

A self-built four-wheel mobile robot with LiDAR mounted on top is used as the experimental platform. The platform is mainly composed of Hokuyo Rapid laser range finder, motor and a wired setup to control the robot. The overall system hardware block diagram is shown in Figure 3.

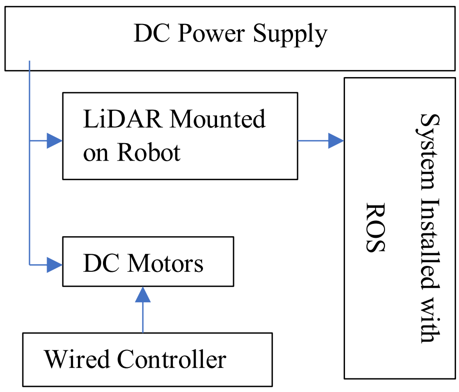

Figure3: Block Diagram of Sys. Hardware

The robot is controlled through a wired setup to move in the environment. The LiDAR is sending the data. To the system which has ROS installed on it.

The system is installed with Ubuntu 18.04 LTS which has ROS Melodic installed on it. DC 12volts motors are being used in the robot to move around in the arena. The robot can be seen in figure4.

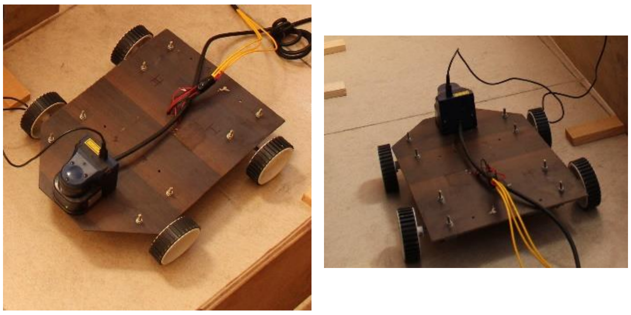

Figure4: Mobile Robot

III. EXPERIMENTS AND RESULTS

The experiment is performed by using ROS (Robot Operating System) on an independent system which is running Linux Ubuntu 18.04 along with the ROS Melodic. After proper installation of ROS, we can now clone the Hector-SLAM GitHub repository into our Catkin workspace and then use it for 2D Mapping.

When the LiDAR is connected to the system, we use the urg_node library to access the sensor data. When urg_node is run it starts to /scan topic which further subscribes to the laser data that is being retrieved from the sensor and starts to publish a 2D Scan of the unknown environment which can be visualized in RViz. But, for Hector-SLAM we need not to visualize the 2D scan, because we have to use the scan directly for the Hector-SLAM.

Now when the Hector-SLAM node is run, it subscribes to the /scan topic and further uses it for making a 2D map of the environment. It also gives the path that was followed by the robot.

As a result, we get a 2D map of the unknown environment and also the path that the robot has moved. We can now save the map using map server and further the map can be used for various purposes such as 2D Navigation, Obstacle avoidance and etc.

The 2D map generated using Hector-SLAM can be seen in figure5(a) and figure5(b).

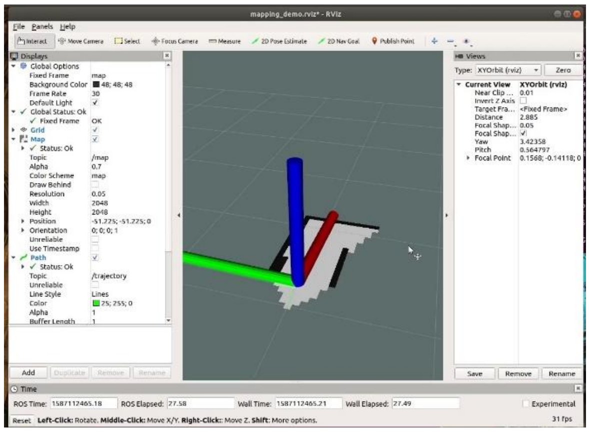

Figure5(a): Starting to map.

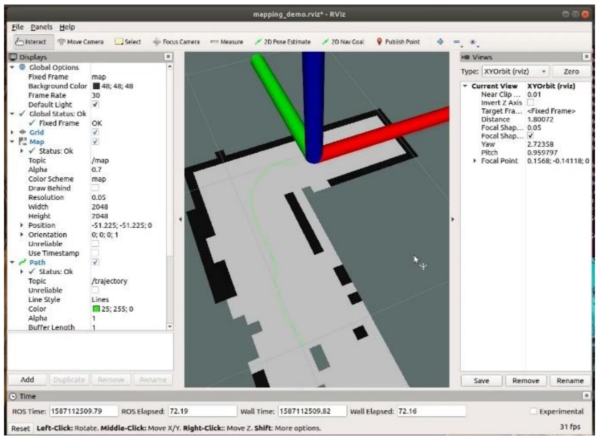

Figure5(b): Ending and completing the map.

It is clearly visible that the robot was moving inside a 'L' shaped environment. The green path is also clearly visible on which the robot has moved.

IV. CONCLUSION

Nowadays mobile robots are in trend. Where ever we can look for automation for solving problems in real life, mostly mobile robots are being used there whether it is a house or a big industrial factory. Mobile robots have always had a hard time taking decisions about the path they may travel to go from an initial position to a final position in an unknown environment either the robot has to pick and place an object or to move a whole rack of components from one room to the other. This paper may conclude that we can use Simultaneous Localization and Mapping (SLAM) for performing all these operations very efficiently. Using Hector-SLAM we can perform mapping in an unknown environment so as to achieve 2D Navigation or to help a mobile robot, take a decision more accurately and easily to choose which path it needs to travel on and even avoid the obstacles which are visible in the map.

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V. REFERENCES

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